

05-01-26

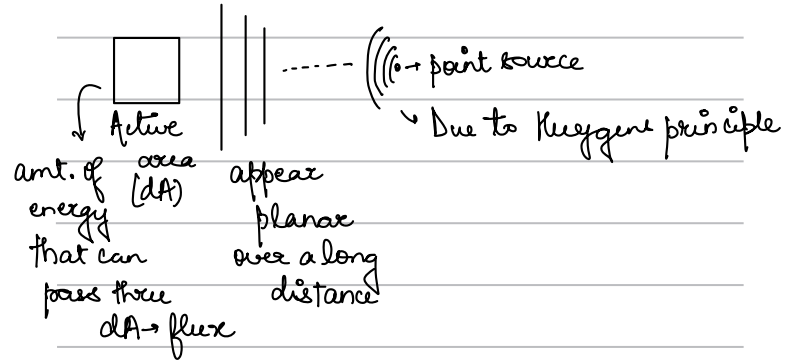
- Quizzes, Assignments, Programming assignments - 30%
- Mid-sem - 30%
- End-sem - 40%

References -

- (1) Essential Radio Astronomy
- (2) Radio Astronomy
- (3) Tools of radio astronomy

- Coherent detection
- Ionosphere - 30 MHz - reflect & send back waves less than these frequencies
- Doppler emission
- CMEs, Plasma, Interplanetary Medium
- Energy emitted by an object - Multimessenger astronomy - neutrinos
- Site Survey - how untouched the EM emission spectrum in that locaⁿ is, the lower it is, the better
- Ground wave | Sky wave communicatⁿ - reflectⁿ from ionosphere - communicatⁿ b/w 1 point to another on Earth - lightning was a problem / disturbance
- Main beam of an antenna - directⁿ & range
- Disturbance from the galactic center - periodic
- Radio five
- Earlier, very noisy amplifiers, too loud - Bell Labs - CMB
- Interferometers, large radars, ALMA, SKA
- Cygnus - one of the 1st quasars to be discovered
- Central engine, particle jets emitted w/ diff. velocities - can be seen only in radio wavelengths

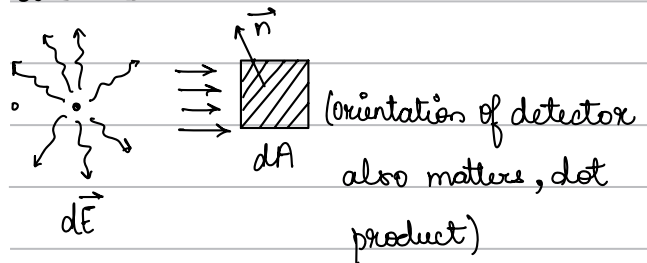
- Synthesis Imaging - CT scans
- Synthetic Aperture Imaging
- Transducer - EM to electrical



- angular extent of the object
- Flux depends on proximity to the source, active area (area of the detector), bandwidth ($d\nu$) of the detector over which the observatⁿ can be made, dt - exposure time

$$\theta = \tan^{-1} \left(\frac{r}{R} \right), \text{ Solid angle } \rightarrow \Omega = \frac{dA}{R^2}$$

- Jet of Cygnus - directⁿal emission, specific intensity - directionality of source taken into consideratⁿ



$$d\vec{E} \propto dA$$

$$\propto dt \text{ (exposure time)}$$

$$\propto d\nu \text{ (bandwidth)}$$

$$\propto d\Omega \text{ (solid angle)}$$

• Optical filters - specified using wavelength
Radio " - " " bandwidth

• Solid angle - small fracⁿ of beam, add'l noise,
need to know true size, error bar

• I_ν - specific intensity

$$d\vec{E} = I_\nu \hat{n} d\vec{A} dt d\nu d\Omega \rightarrow \frac{d\vec{E}}{dt} = I_\nu \hat{n} d\vec{A} d\nu d\Omega$$

↓

$$dP = I_\nu \hat{n} d\vec{A} d\nu d\Omega$$

• Flux density - $\text{Wm}^{-2} \text{Hz}^{-1}$ (object size not properly
known or error, we integrate & lose that term)

• $I_\nu = \frac{dP}{d\nu dA \hat{n} d\Omega}$ [$\text{Wm}^{-2} \text{Hz}^{-1} \text{sr}^{-1}$] → does not change
with distance from the
source

$$= \frac{dP}{d\nu dA \cos\theta d\Omega}$$

- Reciprocity

- $S = \int_\Omega I_\nu d\Omega$, intercepting small portion of the
sphere.

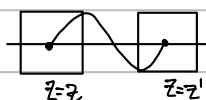
$$\left. \begin{aligned} \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{H} &= \vec{J} + \frac{\partial \vec{D}}{\partial t} \end{aligned} \right\} \text{Maxwell's Eqns}$$

Coherence - Interferometry

• $|A|e^{i\omega t} \rightarrow \cos(\omega t) + j\sin(\omega t)$
 ↳ plane waves - limit to $\vec{E}$ & $\vec{B}$ → orthogonal

Phasor diagram

• Phase of the wave - amt. of angle/degrees that it would have rotated in going from one plane to another → $\Phi = \frac{2\pi}{\lambda} (z - z')$
 ↳ path length



$$\rightarrow A_0 \cos(\omega t - kz)$$

$$\omega t - kz = \text{constant} \rightarrow \frac{d\omega t}{dt} - \frac{dkz}{dt} = 0 \rightarrow \omega = k \frac{dz}{dt}$$

$$- \frac{dz}{dt} = \text{phase velocity } (v_p) = \frac{\omega}{k} \rightarrow \text{wave number}$$

$$k = \frac{2\pi}{\lambda}; \vec{k} = k \cdot \hat{n} \rightarrow \text{fourier components of a signal}$$

Quasi-monochromatic light

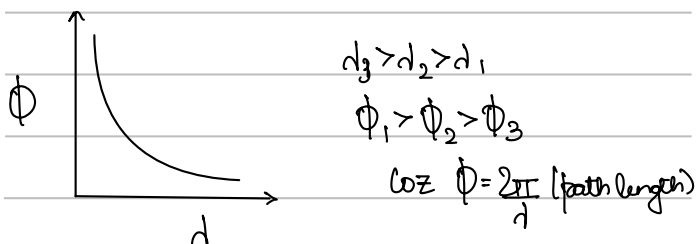
- Passband - half-power points - filter - quality factor - group of waves

• $\omega_1 = \omega_0 - \delta\omega$
 $\omega_2 = \omega_0 + \delta\omega$
 ↳ $k_1 = k_0 - \delta k$
 $k_2 = k_0 + \delta k$

↳ constant $v_p = \frac{\omega}{k}$ - changing k - has to change

↳ bandwidth
 $v_{p,g} = \frac{\delta\omega}{\delta k} = \text{group velocity}$

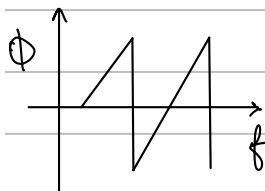
Phase velocity vs Group velocity



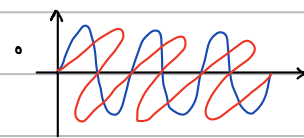
$$\Phi = \frac{2\pi}{\lambda} (\text{path length}) = \frac{2\pi}{c} \cdot f \cdot (\text{path length})$$

$$\Phi_2 - \Phi_1 = \frac{2\pi}{c} \Delta f \cdot L \text{ where } \frac{L}{c} = T_g - \text{group delay}$$

$$T_g = \frac{1}{2\pi} \left(\frac{\Phi_2 - \Phi_1}{f_2 - f_1} \right)$$



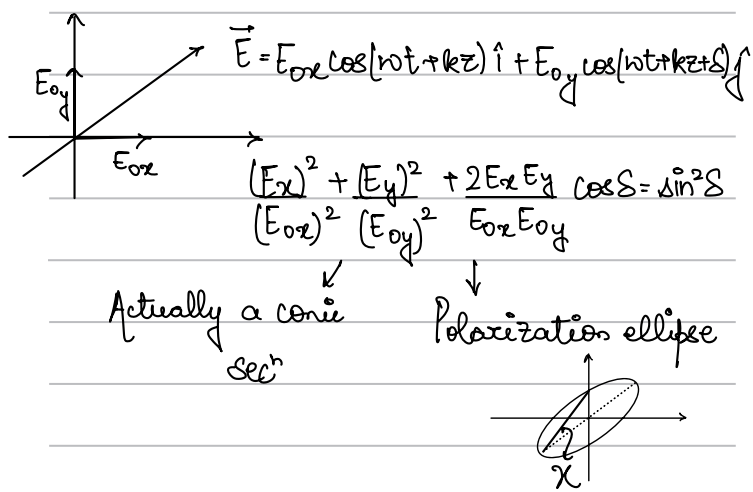
• $\vec{A}e^{j(\omega t - kz)}$ - wave



Polarization
 ↳ Path travelled, disturbances in the path, was there any $\vec{B}$ or not

• Circular polarization - satellite antennas - right
 Oar polarized & left Oar polarized

- δ - Delay in electric field ^{component} oscillating in x from electric field ^{component} oscillating in y



- Vertical signal $\rightarrow$ subscribes an ellipse at the beginning

• When $E_x = E_y$ & in same direcⁿ & no δ , simply a line ($\delta = 0$)

• When $E_x = E_y$ & $\delta = \pi/2$, subscribes an ellipse

• 2 parameters to define it - ψ - polarizaⁿ angle

(Radiopolarimetry) - polarized emission & intermediate medium

$\downarrow$
Faraday Effect

• Polarizaⁿ calibration

• Axial ratio $\rightarrow AR = \frac{a}{b}$

When $AR = 1$ - circular

$AR = \infty$ - linearly polarized

• χ - ellipticity angle $\rightarrow -\frac{\pi}{4} < \chi < \frac{\pi}{4}$ & $\tan(\chi) = \pm \frac{b}{a}$

• $-\pi < \psi < \pi$

• Point Array Sphere

• Coherence - Property of waves that describes how well-correlated they are in phase - maintain a constant phase relationship over time & space - interfere in a predictable & stable way

• Hole in interferometry - Interferometry is the science of overlapping 2 or more waves to extract inf. from the resulting interference pattern

$\rightarrow$ Quality of an interference pattern is called Visibility (V):

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

• Phase velocity - v_p - Speed at which the individual peaks & troughs of a wave move - In some media, can exceed speed of light but does not violate relativity b/c phase alone cannot carry inf.

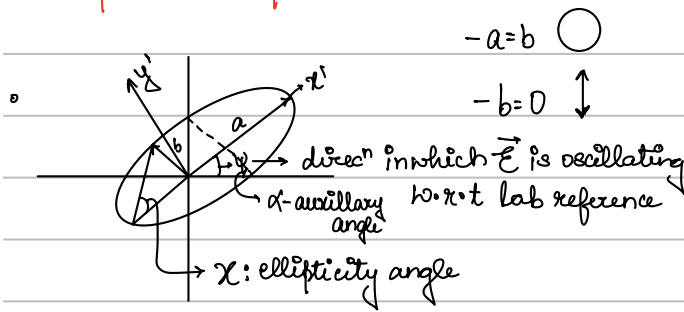
• Group velocity - v_g - Speed at which the overall shape or 'envelope' of the wave packet moves - Speed at which energy & inf. actually travel

• Quasi-monochromatic light - light that has a very narrow range of frequencies ($\Delta \nu$) centered around a mean frequency (ν_0) s.t. bandwidth is much smaller than the mean frequency ($\Delta \nu \ll \nu_0$)

- Finite coherence, amplitude & phase fluctuations

$$E(t) = A(t) \cdot e^{i[2\pi\nu_0 t + \phi(t)]} \quad \text{and} \quad T_c \approx \frac{1}{\Delta \nu}$$

Derive polarizer ellipse



$a^2 + b^2$ - Invariant - Total Intensity Component

4 Stokes parameters $E_x^2 + E_y^2 = I$ $\leftarrow$ 1 Stokes parameter

$a^2 + b^2 = E_x^2 + E_y^2 = \text{Stokes Intensity}$

πab - Invariant - Area of the ellipse

$$\rightarrow a^2 + b^2 = E_{ox}^2 + E_{oy}^2 = I$$

$$\pi ab = \pi E_{ox} E_{oy} \sin \delta \rightarrow \text{what is its physical significance?}$$

$$\tan \alpha = \frac{E_{oy}}{E_{ox}}; \tan \chi = \frac{b}{a}$$

Polarizer Interferometry - multiple readings

$$\begin{bmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{bmatrix} \rightarrow \text{rotation matrix}$$

$$\rightarrow \begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{bmatrix} \begin{bmatrix} E_x' \\ E_y' \end{bmatrix}$$

$$\vec{E} = E_x \hat{i} + E_y \hat{j}$$

Stokes Parameters

4 Stokes parameters

$\rightarrow$ amt. of light polarized in total
1) S_0 - Total Intensity $\rightarrow E_x^2 + E_y^2$

$$S_0 | I = \langle E_x E_x^* \rangle + \langle E_y E_y^* \rangle$$

$\rightarrow$ averaged out over a long time

(2) $S_1 | Q \rightarrow E_x^2 - E_y^2$ (polarized at either 0° or 90°)

(3) $S_2 | U \rightarrow 2 E_x E_y^* \cos \delta \rightarrow S_1 = S_2$ tells us how much is 45° & 135° polarized

(4) $S_3 | V \rightarrow 2 E_x E_y^* \sin \delta \rightarrow$ tells us if it is Only polarized or not
 $\delta_1 - \delta_2 = \frac{\pi}{2} (\text{max})$

$$I^2 \geq Q^2 + U^2 + V^2$$

Degree of Polarization - $p = \frac{\sqrt{U^2 + V^2 + Q^2}}{I}$

$\langle a b \rangle$ - averaged it over a long period of time

Stokes Vector -

$$\begin{bmatrix} I \\ Q \\ U \\ V \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$\downarrow$
right Oax
polarized
or left Oax
polarized

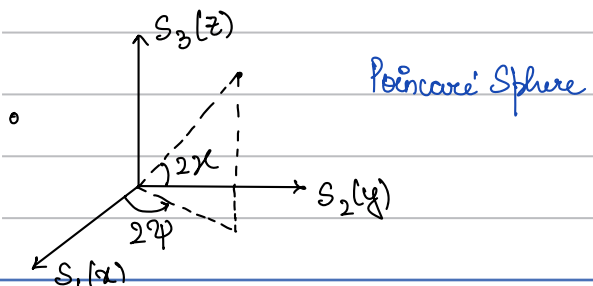
$\begin{bmatrix} E_{ox}^2 + E_{oy}^2 \\ E_{ox}^2 - E_{oy}^2 \\ 2 E_{ox} E_{oy}^* \cos \delta \\ 2 E_{ox} E_{oy}^* \sin \delta \end{bmatrix}$ - Based on ψ & χ , we can write the Stokes parameters

$\rightarrow$ sweep entire hemisphere on one side

$$S_1 = \cos 2\psi \cos 2\chi \text{ (x-comp)}$$

$$S_2 = \sin 2\psi \cos 2\chi \text{ (y-comp)}$$

$$S_3 = \sin 2\chi \text{ (z-comp)}$$



• Faraday Rotation

• Polarization Calibrator

• Quarter wave plates & Half wave plates

• Half-silvered mirror

• Non-depolarizing components

• 4 Stokes parameters-

(1) S_0 or I - Total Intensity - total optical power of the light beam (polarized + unpolarized)

(2) S_1 or Q - Horizontal vs Vertical - difference in intensity b/w horizontal & vertical linear polarization

(3) S_2 or U - $+45^\circ$ vs -45° - difference in intensity b/w linear polarization at $+45^\circ$ & -45°

(4) S_3 or V - Right vs Left Oax - difference in intensity b/w Right-Hand & Left-Hand Oax polarization

• If Stokes vector = $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$ - light is completely right

Only polarized, if it were -1 instead of 1 at the bottom - light would have been left Only polarized - $\vec{E}$ rotates in a perfect $\odot$ in a clockwise direction (when looking towards the source) as the wave travels thru space - This is the state used most often for GPS & satcom to avoid orientation issues

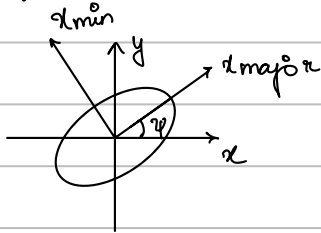
• If Stokes vector = $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ - light has no preferred direction

of vibration - $\vec{E}$ oscillates in all possible directions (horizontal, vertical, Oax etc) randomly & rapidly - if you put any single polarizer (linear or Oax) in front of this beam, exactly 50% of the light will always pass thru, regardless of how you rotate the polarizer

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◦ Polarization Diversity

- Manipulate polarization - adjust phases or rotate the plane of polarization



$$\text{Rotation} \rightarrow \begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} \cos \psi & -\sin \psi \\ +\sin \psi & \cos \psi \end{bmatrix} \begin{bmatrix} E_{x \max} \\ E_{x \min} \end{bmatrix}$$

Jones Matrix

ψ - polarization angle

$$E_x = E_{0x} \sin(\omega t - kz + \delta_x)$$

$$\begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} E_{0x} e^{-i\delta_x} \\ E_{0y} e^{-i\delta_y} \end{bmatrix}$$

$$E_y = E_{0y} \sin(\omega t - kz + \delta_y)$$

Jones Vector

$$\vec{E}_{\text{new}} = J \cdot \vec{E}_{\text{old}} \rightarrow X = \begin{bmatrix} 1 \\ 0 \end{bmatrix}; Y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}; E_{45^\circ} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$E_{\alpha} = \begin{bmatrix} 1 \\ \pm i \end{bmatrix}$$

$$X \rightarrow E_{45^\circ} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$X \rightarrow E_{\alpha} = \begin{bmatrix} 1 & 0 \\ i & 0 \end{bmatrix} \rightarrow \text{Quarter-wave plate}$$

◦ Quadrature Hybrid

$$J_{ADC} J_{\text{amp}} J_{QWP} J_{\text{rot}} \begin{bmatrix} E_{\text{inc}x} \\ E_{\text{inc}y} \end{bmatrix} = \vec{E}_x$$

Read more about this incidence matrix

◦ Müller Matrices

$$S = \begin{bmatrix} I \\ Q \\ U \\ V \end{bmatrix} \rightarrow S_{\text{new}} = M S_0$$

4x4 Müller matrix

$$\begin{bmatrix} J_1 \otimes J_2 \end{bmatrix} \rightarrow \begin{bmatrix} m_{00} & \dots & m_{03} \\ \vdots & & \vdots \\ m_{30} & \dots & m_{33} \end{bmatrix}$$

for $\vec{E}$ for intensities
OR $J_1 \otimes J_1$
averages of these intensities

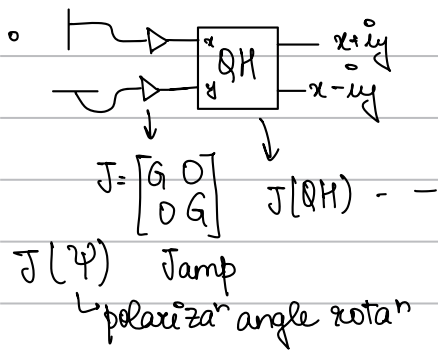
◦ Radiative Transfer

$$dI_0 = -k I_0 dl$$

thickness of layer
absorption by layer

depends on wavelength used for observaⁿ, spatial independence

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Radiative Transfer

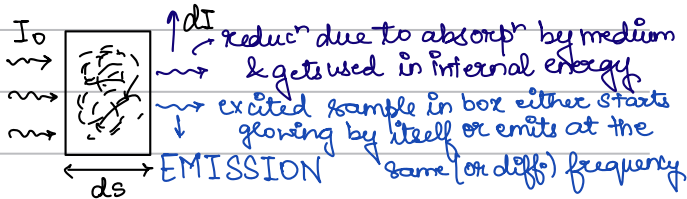
- Atmosphere - neutral medium

But many layers which have free e's & -ve charge

• Intergalactic + Interstellar + Interplanetary Mediums

↓
Plasma

ABSORPTION



• Scattering - of the order of the wavelength itself (the medium)

$-dI \propto I_0 \cdot ds \rightarrow dI = k I_0 ds$

↓
absorpⁿ coefficient

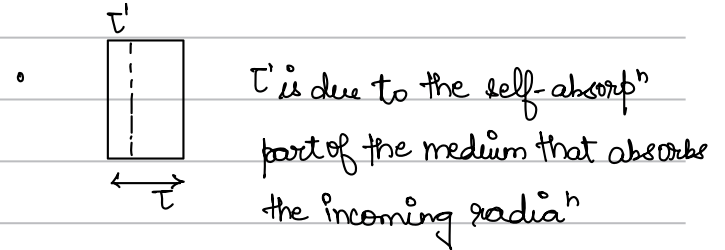
• Shortest wavelength IR - penetrates thru fog completely

$\tau = \int_0^L k ds$

↓
optical depth

$\int \frac{dI}{I} = -\int \tau \rightarrow I = I_0 e^{-\tau} \rightarrow \text{Beer Lambert's law}$

$I = I_0 e^{-kL}$ where k is constant



$dI = j \cdot ds e^{-(\tau - \tau')}$

↓
emissivity

$I = \int j \cdot ds e^{-(\tau - \tau')}$

$I_\nu = \int_0^L \frac{j_\nu}{k} [1 - e^{-\tau}]$

↓
 S_ν - source funcⁿ

Read more about this

$I = I_0 e^{-\tau} + S [1 - e^{-\tau}]$

• Photosphere of a star - Surface where $\tau = 1$

• Water Vapour Radiometry

• What about in case of e's & ions in medium?

$A e^{i(\omega t - k z)}$ propagator term

$e^- p^+ \quad p^- e^-$

$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

$\nabla \times \vec{H} = \vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t}$

$\nabla \times \vec{B} = \vec{J} \mu + \mu \epsilon_0 \frac{\partial \vec{E}}{\partial t}$

↓
 $\frac{\vec{B}}{\mu}$

$$\frac{\partial \vec{E}}{\partial t} = i\omega A e^{i(\omega t - kz)}$$

$$= i\omega \vec{E}$$

$$\frac{\partial^2 \vec{E}}{\partial t^2} = -\omega^2 \vec{E}$$

$$\frac{\partial \vec{E}}{\partial z} = -ik A e^{i(\omega t - kz)} = -ik \vec{E}$$

$$\frac{\partial^2 \vec{E}}{\partial z^2} = k^2 A e^{i(\omega t - kz)} = -k^2 \vec{E}$$

$$\nabla \times \vec{E} = \frac{\partial}{\partial t} \left(\vec{J} \mu_0 + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

Replace $\vec{J}$ with $n_e \cdot e \cdot \vec{v}$ - velocity

$$\text{Losses} - dI_{\nu, \text{loss}} = -\alpha_{\nu} I_{\nu} ds$$

$$\text{Gains} - dI_{\nu, \text{gain}} = f_{\nu} ds$$

$$\frac{dI_{\nu}}{ds} = f_{\nu} - \alpha_{\nu} I_{\nu}$$

$$dI_{\nu} = \alpha_{\nu} ds \quad \& \quad S_{\nu} = f_{\nu} \rightarrow \frac{dI_{\nu}}{d\tau_{\nu}} = S_{\nu} - I_{\nu}$$

$$e^{\tau_{\nu}} \frac{dI_{\nu}}{d\tau_{\nu}} + I_{\nu} e^{\tau_{\nu}} = S_{\nu} e^{\tau_{\nu}} \rightarrow \frac{d}{d\tau_{\nu}} (I_{\nu} e^{\tau_{\nu}}) = S_{\nu} e^{\tau_{\nu}}$$

$$\int_{I_0 e^0}^{I_{\nu} e^{\tau_{\nu}}} d(I_{\nu} e^{\tau_{\nu}}) = \int_0^{\tau} S_{\nu} e^{\tau_{\nu}} d\tau_{\nu} \rightarrow I_{\nu} e^{\tau} - I_0 = S_{\nu} [e^{\tau} - 1]$$

$$= S_{\nu} (e^{\tau} - 1)$$

$$I_{\nu} = I_0 e^{-\tau} + S_{\nu} (1 - e^{-\tau})$$

→ Derivaⁿ of I_{ν} in terms of
Beer-Lambert + S_{ν}

• Derivation of eqⁿ of polarizaⁿ ellipse

$$E_x(t) = E_{0x} \cos(\omega t)$$

$$E_y(t) = E_{0y} \cos(\omega t + \delta)$$

$$\frac{E_y}{E_{0y}} = \cos \omega t \cos \delta - \sin \omega t \sin \delta$$

$$\cos \omega t = \frac{E_x}{E_{0x}} \rightarrow \sin \omega t = \sqrt{1 - \left(\frac{E_x}{E_{0x}}\right)^2}$$

$$\frac{E_y}{E_{0y}} = \frac{E_x}{E_{0x}} \cos \delta - \sqrt{1 - \left(\frac{E_x}{E_{0x}}\right)^2} \sin \delta$$

$$\frac{E_y}{E_{0y}} - \frac{E_x}{E_{0x}} \cos \delta = - \sqrt{1 - \left(\frac{E_x}{E_{0x}}\right)^2} \sin \delta$$

$$\left(\frac{E_y}{E_{0y}}\right)^2 + \left(\frac{E_x}{E_{0x}}\right)^2 \cos^2 \delta - \frac{2E_x E_y \cos \delta}{E_{0x} E_{0y}} = \sin^2 \delta - \left(\frac{E_x}{E_{0x}}\right)^2 \sin^2 \delta$$

$$\frac{E_x^2}{E_{0x}^2} + \frac{E_y^2}{E_{0y}^2} - \frac{2E_x E_y \cos \delta}{E_{0x} E_{0y}} = \sin^2 \delta$$

• Dispersion Measure

Ch-2, Ch-

• Rotation Measure - Tools of Radio Astronomy

• Blackbody Radiation / Rayleigh Jeans Limit

- Spectral distribution of a system in thermal equilibrium
→ amount of radiation emitted / area / time / freq / solid angle

$$B_T(\nu) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1} \quad \text{Blackbody Radiation}$$

$$|B_T(\nu) d\nu| = |B_T(\lambda) d\lambda|$$

↓

$$B_T(\lambda) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda k_B T}} - 1}$$

• Total brightness - $B(T) = \frac{2h}{c^2} \int_0^\infty \frac{\nu^3}{e^{h\nu/k_B T} - 1} d\nu$

$$= \frac{2\pi^4 k^4}{15c^2 h^3} T^4$$

Stefan Boltzmann → $B_{\text{total}}(T) = \sigma T^4$

$$\sigma = 1.8047 \times 10^{-5} \frac{\text{erg}}{\text{cm}^2 \text{s}} \text{K}^{-4}$$

• $B(\nu)$ - Amount of energy radiated per unit area per unit time per unit freq. interval per unit solid angle

$$E = |B(\nu) d\nu| dt dA d\Omega$$

$$= |B(\lambda) d\lambda| dt dA d\Omega$$

↳ this is the invariant here - energy radiated

While computing power, take care of the $(d\nu)$ factor

• Rayleigh Jeans Limit

$$\frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1} \rightarrow \text{If } \frac{h\nu}{k_B T} \ll 1$$

$$e^{\frac{h\nu}{k_B T} - 1} \approx \frac{h\nu}{k_B T}$$

$$B_{RJ}(\nu) \approx \frac{2h\nu^3}{c^2} \frac{1}{\frac{h\nu}{k_B T}} = \frac{2\nu^2 k_B T}{c^2}$$

↓

$$B_{RJ}(\nu) \propto T$$

$\int_{-\infty}^{\infty} B(\nu) d\nu$ → infinite energy from a source not feasible
↓
Planck's constant

• Validity of the RJ limit

$$h = 6.63 \times 10^{-34} \text{ m}^2 \text{ kg} / \text{s}$$

$$k = 1.38 \times 10^{-23} \text{ m}^2 \text{ kg} / \text{s}^2 \text{ K}^{-1}$$

$$\frac{h\nu}{k_B T} \ll 1 \rightarrow \nu \ll T \left(\frac{k}{h} \right) \ll 2 \times 10^{10} \text{ Hz K}$$

$$\frac{\nu}{\text{GHz}} \ll \frac{20T}{\text{K}}$$

$$B = \frac{2\nu^2 k_B T}{c^2} = I_\nu$$

• Brightness Temperature $T_B = \frac{(I_\nu) c^2}{2\nu^2 k}$

Flux density - $S_\nu = I_\nu \Delta\Omega$

- Model the source as a Gaussian profile

$$\left(\frac{S_p}{J_y}\right) = 2 \cdot 65 T \left(\frac{\theta}{\text{arc min}} \right)^2 \left(\frac{\lambda}{\text{cm}} \right)^{-2}$$

$\downarrow$
 $(\theta_{\min} \theta_{\max})$

- θ - smaller, T - higher, source considered to be brighter

• Sometimes the RJ limit is applied even when

$$\left(\frac{\nu}{\text{GHz}}\right) \ll \left(\frac{20T}{\text{K}}\right) \rightarrow \left(\frac{\text{turn}}{\text{unit}}\right) \text{ format}$$

$\downarrow$

In this case, radiation temperature but not the thermodynamic temperature of a source

• For propagation through a medium with optical depth τ -

$$T_b(s) = T_b(0) e^{-\tau(s)} + \int_0^{\tau(s)} T(s) e^{-\tau} d\tau$$

$\swarrow$ Source (absorption) $\swarrow$ Background emission $\swarrow$ medium radiating emission
 s - distance along line of sight

• Consider a situation with no source, only medium is emitting $\rightarrow T_b(0) = 0$ & the medium is isothermal (T_{medium}) - medium is uniformly heated along LOS

$\downarrow$

For isothermal

$$T_b(s) = T_b(0) e^{-\tau(s)} + T_{\text{med}} (1 - e^{-\tau(s)})$$

• For no source - $T_b(s) = T_{\text{med}} (1 - e^{-\tau(s)})$

• Limit 1 - $\tau_y \ll 1 \rightarrow$ optically thin

$$T_b(s) \approx T_{\text{med}} \tau_y$$

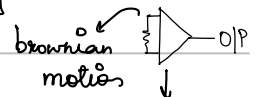
• Limit 2 - Optically thick $\tau_y \gg 1$

$$T_b = T_{\text{med}}$$

• What are we calling temperature here??

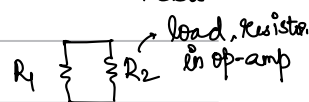
• Noise of a 'warm' resistor -

- The e^- in the resistor undergo 'thermal motions' giving rise to a current



$$\langle i \rangle_{\text{over long enough time}} = 0 \text{ but } \langle i^2 \rangle \neq 0$$

Maxwell's Distribution



- Gives rise to a noise - Johnson noise

$\rightarrow$ noise peaks when impedance is matched

• Average power per unit bandwidth -

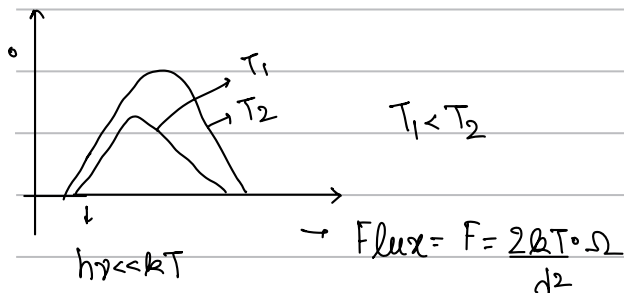
$$P_{\nu} = \langle i^2 V \rangle = \frac{\langle V^2 \rangle}{2R} = kT$$

$\nwarrow$ voltage

23-01-26

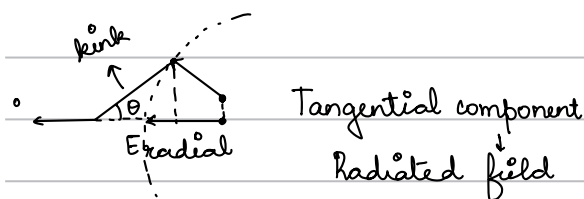
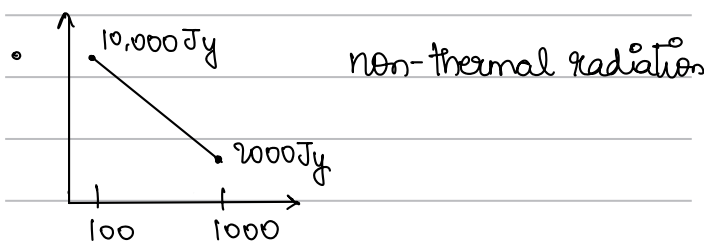
• $B_\nu(T) = \frac{2kT}{\lambda^2}$ → Brightness temperature

$B_\nu(T) = \frac{2kT\nu^2}{c^2}$

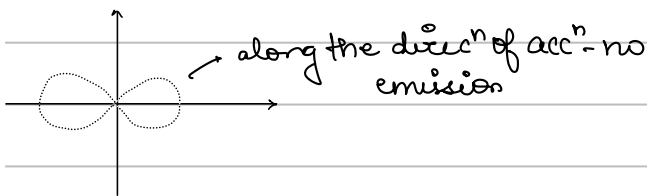


• $F = \int B \cdot d\Omega$

• Example - numerical



• $E_{rad} \propto \frac{a \sin \theta}{r} \rightarrow E_{rad} = \frac{ca \sin \theta}{r} ; P \propto \frac{a^2 \sin^2 \theta}{r^2}$



• If $a = \sin \omega t$, field would also vary harmonically

• $P = \frac{qa^2}{6\pi c^3 \epsilon_0}$ → Larmor formula

• $T \sim 10^6$ K - temperature at which solar corona exists - coulombic interact

• Bremsstrahlung emission → free-free

• Essential Radio Astronomy

• $j^{\text{ff}} \propto n_i n_e (T)^{1/2} e^{-h\nu/kT}$

• $\frac{j^{\text{ff}}}{\alpha^{\text{ff}}} \approx B_\nu(T) \rightarrow \frac{j^{\text{ff}}}{\alpha^{\text{ff}}} \approx \frac{2kT\nu^2}{c^2}$

$\alpha^{\text{ff}} \propto n_i n_e T^{-3/2} \nu^{-2}$

• More absorpⁿ at low frequencies - emission is optically thick

•
$$\begin{aligned} & e^{-j^{\text{ff}}(\omega t - kd/2)} + e^{-j^{\text{ff}}(\omega t + kd/2)} \\ & e^{-j^{\text{ff}}\omega t} [e^{-kd/2} + e^{kd/2}] \\ & \downarrow \\ & -d/2 \quad +d/2 \\ & E \sim N^2 \\ & \text{Larger no. of } e^- \text{ s} \\ & \text{getting acc. harmonically} \\ & P \sim N^4 \end{aligned}$$

• Coherent emission

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• Thermal Bremsstrahlung or free free emission

$$-j_{\nu}^{ff}, \alpha_{\nu}^{ff} - \frac{j_{\nu}^0}{\alpha_{\nu}} = S_{\nu} \xrightarrow{\text{source func}^n} \propto T_b; \frac{2kT}{d^2} \text{ or } \frac{2k\nu^2}{c^2}$$

$$-j_{\nu}^{ff} \propto n_e n_i T^{-1/2}; \alpha_{\nu}^{ff}$$

$$S_{\nu} = \frac{2k\nu^2}{c^2} \Delta\Omega T_b$$

$$S_{\nu} = I_{\nu} \times \Delta\Omega$$

$$P = I_{\nu} \times \Delta\Omega \times A \times B$$

$$\bullet T = \int \alpha ds$$

↓
absorptivity

$$\bullet I_{\nu} = I_0 e^{-\tau} + B_{\nu}(1 - e^{-\tau})$$

↓
source

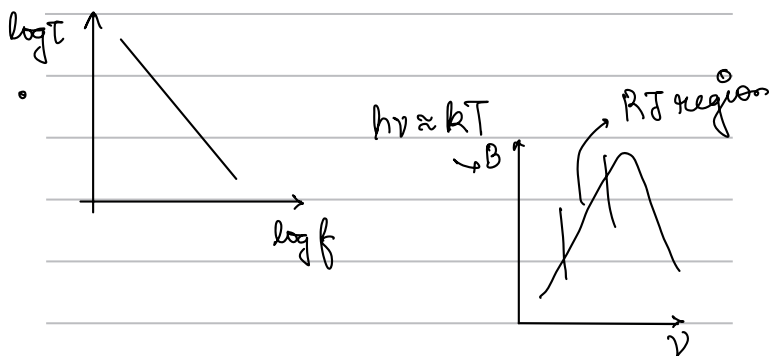
• n_e - much higher when closer to the Sun
& reduces as we go farther

$$\tau \propto \nu^{-2} \int n_e n_i ds \quad \text{— emission measure}$$

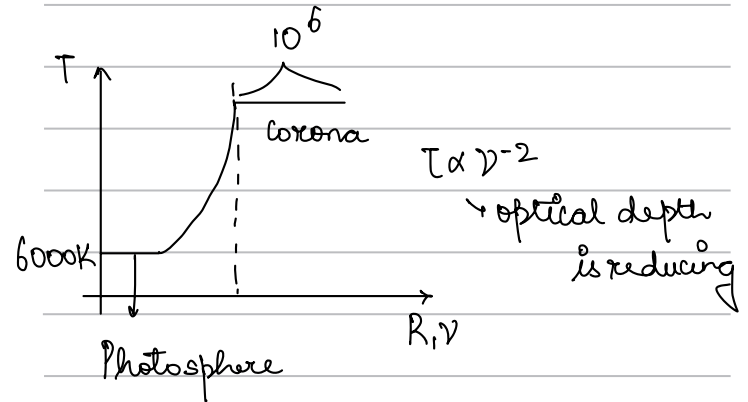
$$\tau \propto T^{-3/2} \nu^{-2} \int n_e n_i ds$$

• $\tau \uparrow$ - medium opaque

$\tau \downarrow$ - medium transparent



$$\bullet B_{\nu}(T) = \frac{2k\nu^3}{c^2} \times \frac{1}{e^{h\nu/kT} - 1}$$



• Frequency at which we observe tells us about the epicenter of observation emission

• Multi-frequency imaging

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$$\bullet B(T) = \frac{2kT}{\lambda^2} \rightarrow 10^{11-12} K$$

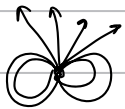
• Non-thermal emission mechanisms -

• An accelerated e^- emits radiation

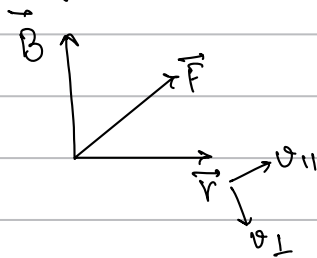
→ mildly relativistic

(1) Cyclotron emission

(2) Synchrotron emission [$v_e \approx c$]



- electron travelling at mildly relativistic speed, gets captured in the $\vec{B}$



• Drift velocity



• Not experience any Coulombic force from other electrons

$$\vec{F} = q(\vec{v} \times \vec{B}) \rightarrow qvB \quad \text{max when } \vec{v} \perp \vec{B}$$

$$\bullet \frac{m_e v^2}{r} = qvB \rightarrow \omega = \frac{qB}{m_e}$$

angular velocity - rate at which the e^- is rotating around the $\vec{B}$

$$\omega = \left(\frac{e}{m_e} \right) B \rightarrow f_{cyc} = \frac{1}{2\pi} \left(\frac{e}{m_e} \right) B$$

↑
cyclotron frequency

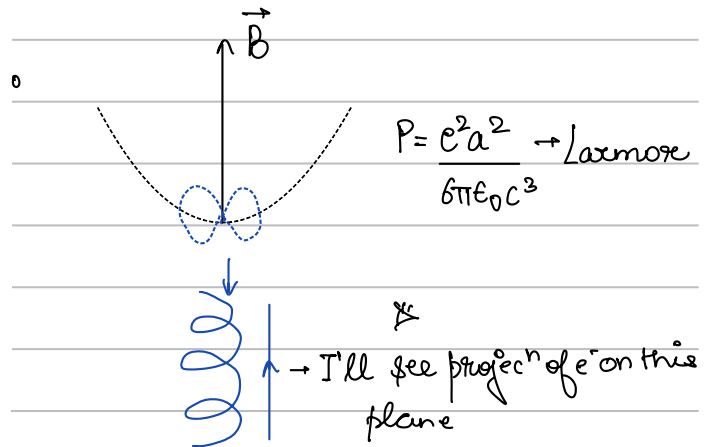
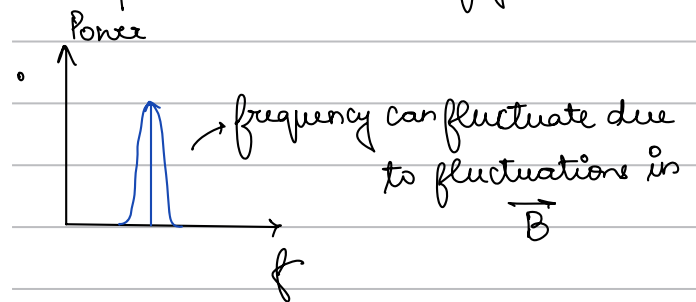
2.8×10^{10}

$f_{cyc} \propto B$

- Unless $\vec{B}$ changes, f_{cyc} cannot change - higher velocity e^- has to go thru a larger spiral to conserve f_{cyc}

$$\text{- Earth's } \vec{B} = 65 \mu T \rightarrow f_{cyc} \sim 10^5 \text{ Hz}$$

$$\text{Sunspots - } 10,000 \text{ Gauss} \rightarrow f_{cyc} \sim 10^{10} \text{ Hz}$$



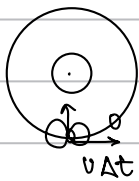
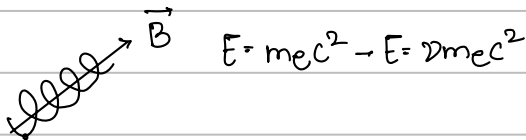
$$P = \frac{e^4 v_{\perp}^2 B^2}{6\pi\epsilon_0 c^3 m_e^2} = \frac{e^2}{6\pi\epsilon_0 c^3} \frac{\omega^2 v_{\perp}^2}{(\alpha B)^2}$$

- v does not decide frequency but decides power emitted / radiated

• e^- travelling in stronger $\vec{B}$ emits more power than the one travelling in weaker $\vec{B}$

• P measured in general - not constant - group of e^- s - same f_{cyc} but incoherent emission of P

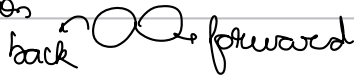
• Synchrotron ($v \perp c$)



The emission would have travelled a distance of $c\Delta t$ - distance travelled

by emission & e^- would be $c\Delta t$

• e^- is travelling at such high speeds that forward emission is $(c-v)\Delta t$ from the backward emission



• High speeds - Tunnel vision - decrease in field of view $\rightarrow$ Lorentz factor

$$\frac{1}{\sqrt{1-v^2/c^2}}$$

• $P(\theta) \propto \cos(\Phi)$

$\downarrow$

$$\Phi = \frac{2\pi}{\lambda} \times \text{path length}$$

Power increase - reduction in region where constructive interference can happen



distance b/w these 2 points is so huge in space
acts as a point source

acts as 2 diff.

point sources in 2 diff. loc's

combine as if 2 point sources superposing in

the far field - same wavefront but a phase difference - kind of like Doppler Effect

• Length contraction

$$\text{measured length} = \frac{L}{\gamma} = \frac{L}{\sqrt{1-v^2/c^2}}$$

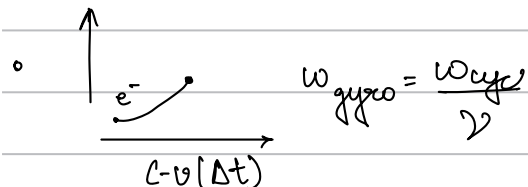
$$\omega_{gyro} = \frac{eB}{\gamma m_e}$$

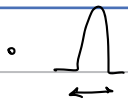
$\downarrow$

$\omega_{gyro} < \omega_{cyc}$ for a given $\vec{B}$ b/c of

the γ factor

• When the observer is seeing, it has also reduced - as e^- is moving towards observer, causes a blueshift





width of pulse $\rightarrow$

$$\Delta t_{\text{pulse}} \propto \underbrace{\left(1 - \frac{v}{c}\right)}_{\frac{1}{2\gamma^2}} \Delta t_{\text{beam}}$$

$$\Delta t_{\text{pulse}} = \frac{1}{2\gamma^2} \Delta t_{\text{beam}}$$

pulse in the time domain

$\rightarrow$ In fourier domain - power
contained in many frequencies - 1 across the
whole fourier domain